Department of Electrical & Electronic Engineering
Brac University
Semester: Spring 25

EEE101L

ELECTRICAL CIRCUITS I LABORATORY

Section:05

Hardware Experiment 06+07

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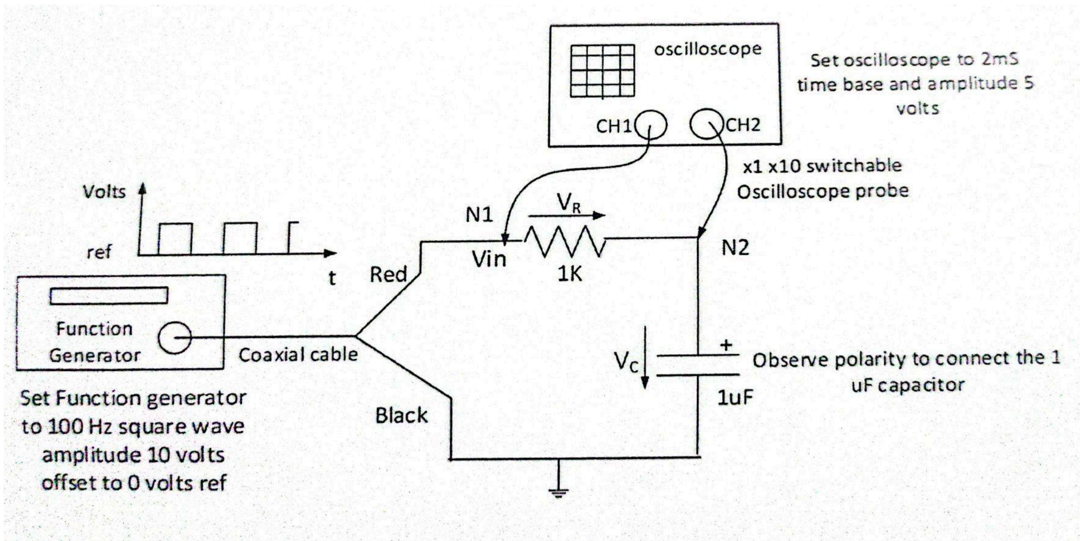
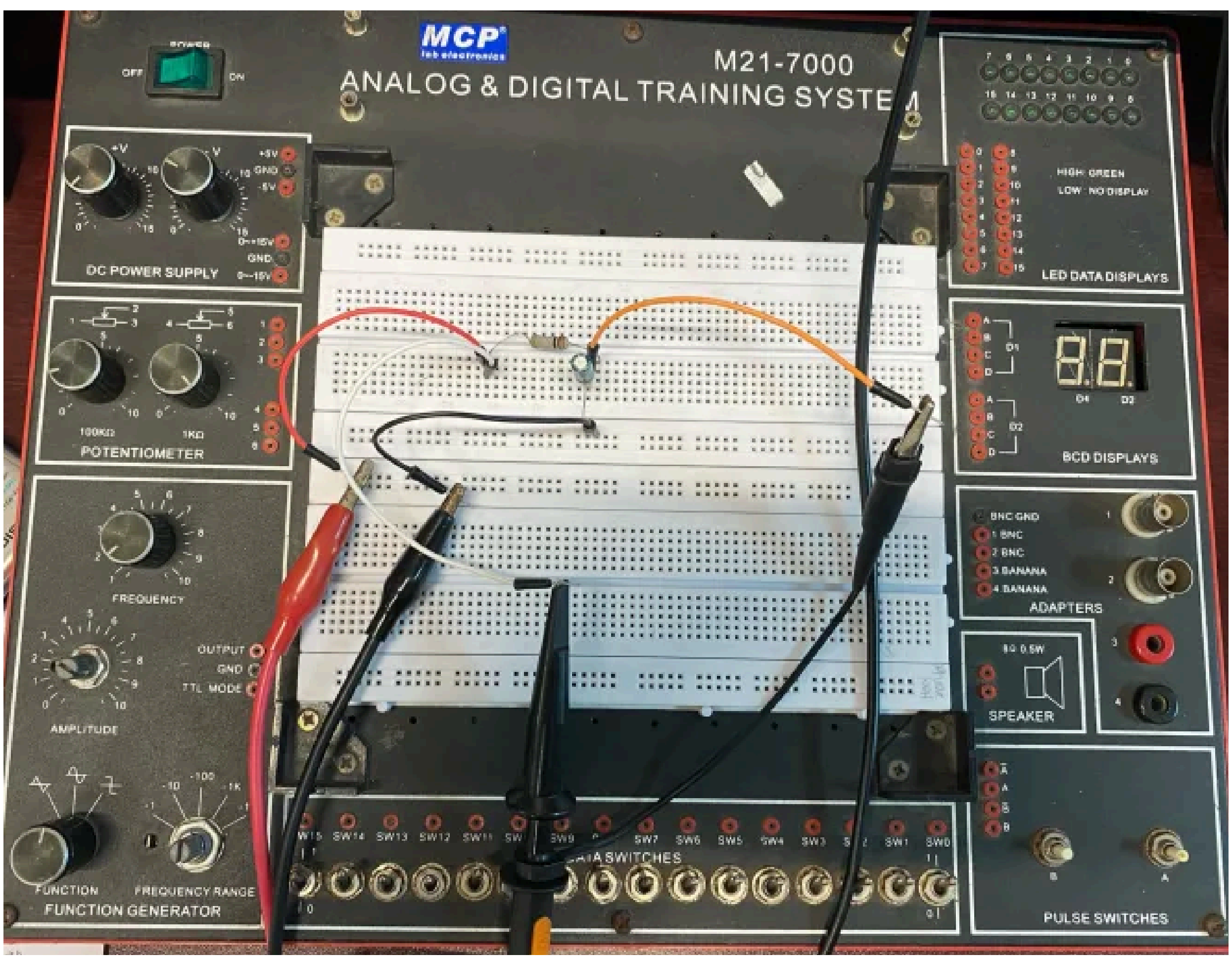
Objective: The Objective of this experiment is to get introduced to the ac voltage and the oscilloscope.

Experimental Setup:

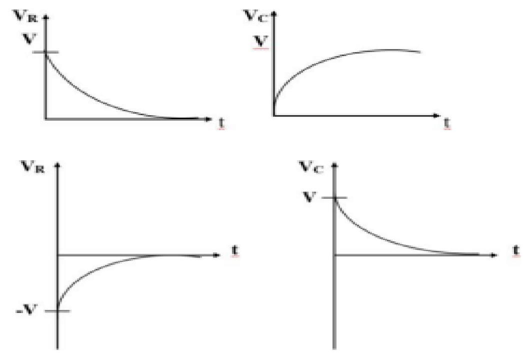


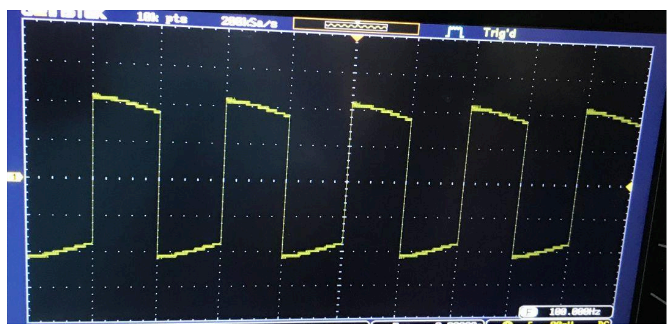
Experimental setup to observe transient behaviour of a RC circuit

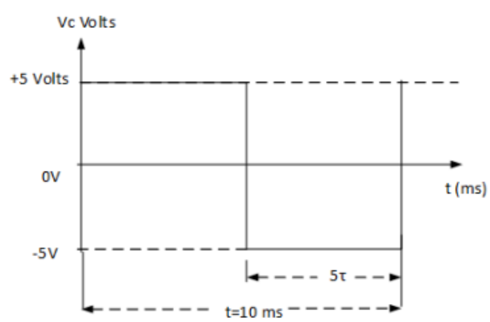
Fig. 1



Waveforms:

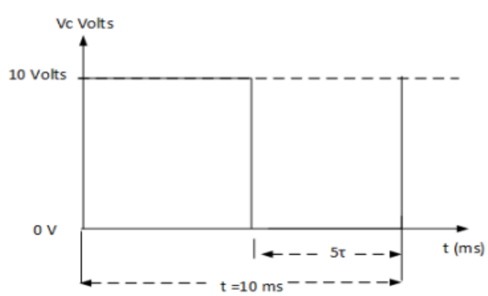




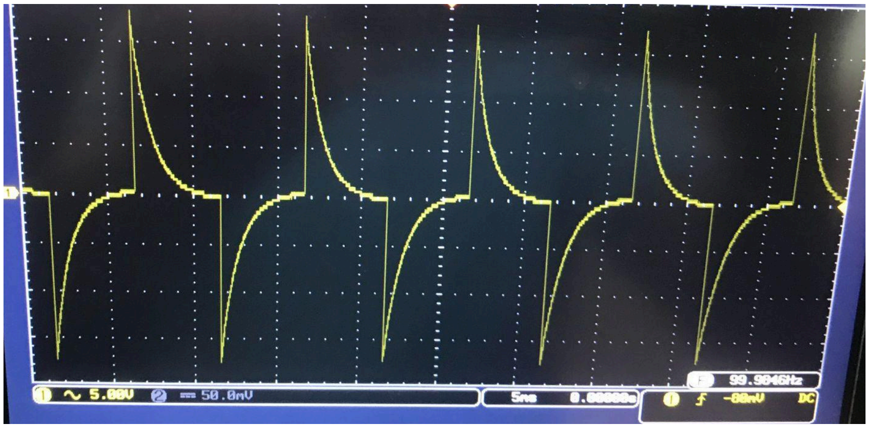


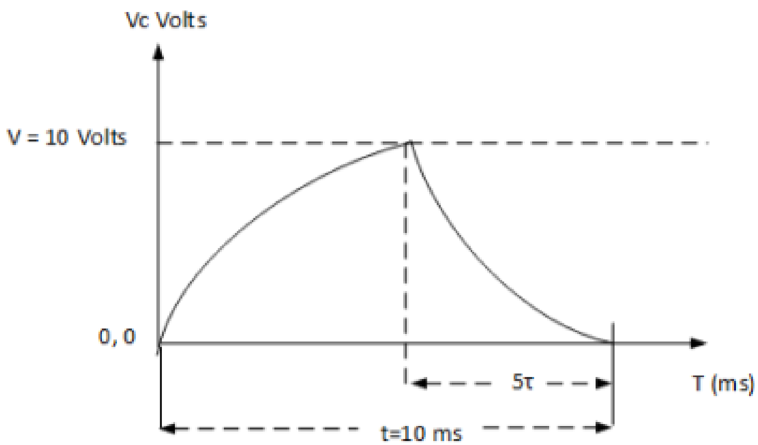
An NRZ Square wave with frequency of 100 Hz with amplitude +5 Volts – 5 Volts volts along the reference axis

τ = 1mS

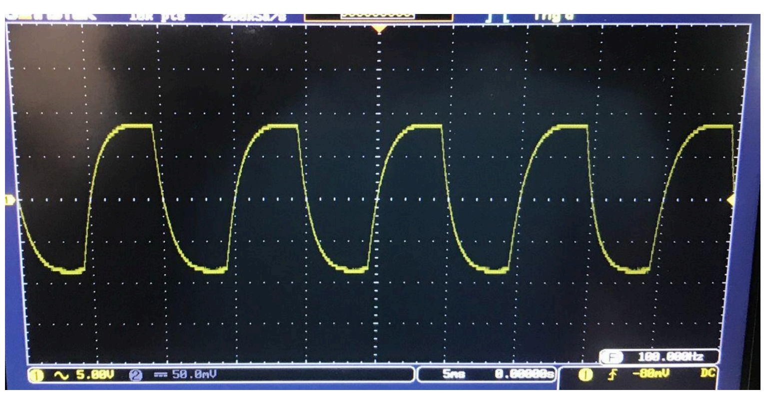


Square wave = 100 Hz with amplitude 10 volts offset adjusted above the reference axis





Waveform showing charging and discharging of an RC circuit



Question:
Define time constant for an RC circuit. What is the significance of time constant? How time constant can be determined?

Answer:
The time constant (τ) of an RC circuit is a measure of how quickly the voltage across the capacitor charges or discharges through the resistor.

$$\tau = R \times C$$

The time constant indicates how quickly the capacitor charges or discharges in response to a voltage change. A smaller time constant means a faster response, while a larger one means the circuit reacts more slowly. After about 5 time constants, the capacitor is considered fully charged or discharged. This is useful in timing circuits, filters, and signal processing applications.

The time constant can be calculated using the values of resistance and capacitance ($\tau = RC$). Alternatively, it can be found experimentally by applying a step voltage and measuring the time it takes for the capacitor's voltage to reach 63.2% of its final value.

Appendix:

Electrical Circuits I Laboratory

EEE 101L

Department of Electrical & Electronic Engineering (EEE)

Brac University

Experiment No. 6 +7

Generation and Measurement of Waveforms

1. **Objective:** The objective of this experiment is to get introduced to the ac voltage source and the oscilloscope.

2. **Theoretical Background:**

AC Voltage Source

It generates an ac voltage.

It can deliver 3 waveforms. They are square, triangular and sinusoidal.

The frequency of the voltage can be set up by selecting a frequency range and turning the frequency knob.

The amplitude of the voltage can be set up by turning the amplitude knob.

A dc voltage can be added to the ac voltage by pulling out the offset knob and turning it.

Oscilloscope

It shows the graph of a voltage. The x axis shows time and the y axis shows voltage.

The screen – It looks like a graph paper. The horizontal line at the middle is the x axis. The vertical line at the middle is the y axis.